DC Machines — Exam Quick Notes

Construction | DC Generator | DC Motor

Q1 — Construction of DC Machine (10 Marks)

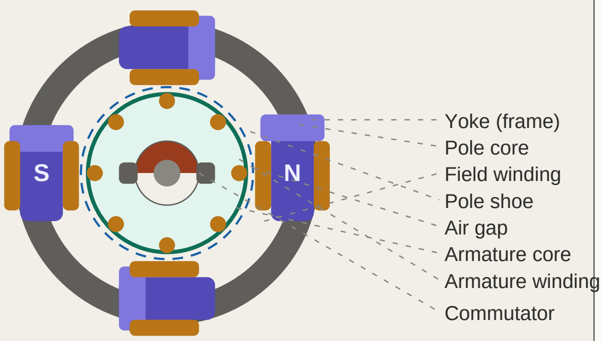

Fig 1: DC machine cross-section (4-pole)

Parts to write (1 mark each):

- **Yoke:** Cast iron / rolled steel — frame + flux path
- **Pole core:** Silicon steel laminations — carries field winding
- **Pole shoe:** Widened end of pole — spreads flux uniformly
- **Field winding:** Copper coils on pole — creates magnetic flux
- **Armature core:** Laminated Si-steel — reduces eddy current losses
- **Armature winding:** Copper conductors — EMF / torque produced here
- **Commutator:** Cu segments + mica — converts AC to DC output
- **Brushes:** Carbon — collect current from commutator

Key: Label diagram clearly with all 8 parts for full marks

Important Formulas

EMF Equation (Generator):

E = (P x phi x N x Z) / (60 x A)

P = poles, phi = flux/pole, N = speed (rpm), Z = conductors, A = parallel paths

Torque Equation (Motor):

T = (P x phi x Z x Ia) / (2 x pi x A)   **T proportional to phi x Ia**

Back EMF: Eb = V - Ia x Ra (self-regulating mechanism of motor)

Fleming's Rules — Must Remember!

DC GENERATOR

Rule: Fleming's RIGHT-Hand Rule
Thumb --> Direction of Motion
Index --> Magnetic Field (B)
Middle --> Induced EMF direction

DC MOTOR

Rule: Fleming's LEFT-Hand Rule
Thumb --> Direction of Force (Motion)
Index --> Magnetic Field (B)
Middle --> Current direction (I)

Motor vs Generator — Quick Comparison

Feature	DC Motor	DC Generator
Energy conversion	Electrical --> Mechanical	Mechanical --> Electrical
Input	DC voltage supply	Mechanical rotation (prime mover)
Working principle	F = B I L (force on conductor)	e = B l v (EMF in conductor)
Fleming's rule	Left-Hand Rule	Right-Hand Rule
Back EMF	Present — self-regulating	Not applicable
Commutator role	Maintains unidirectional torque	Converts AC to DC at output

Working Principles — Diagrams & Key Points

Q2 — Working Principle of DC Generator (8–10 Marks)

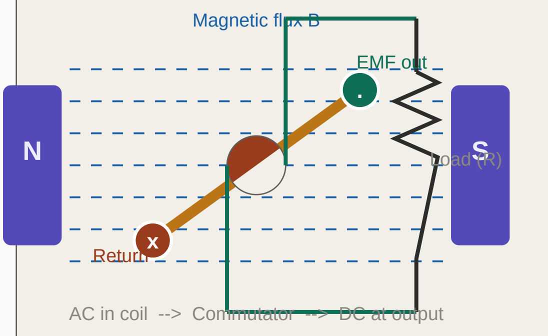

Fig 2: DC Generator — coil rotating in field

What to write:

- Principle: Faraday's Law — conductor cuts flux = EMF induced
- Rule: Fleming's RIGHT-Hand Rule gives direction of EMF
- Armature internally generates AC (sinusoidal)
- EMF = MAX when coil parallel to field (max flux cutting)
- EMF = ZERO when coil perpendicular to field (no cutting)
- Commutator = mechanical rectifier, converts AC to DC
- More coils = smoother DC output (less ripple)
- Formula: E = P x phi x N x Z / (60 x A)

E = P x phi x N x Z / (60 x A)

Q3 — Working Principle of DC Motor (8–10 Marks)

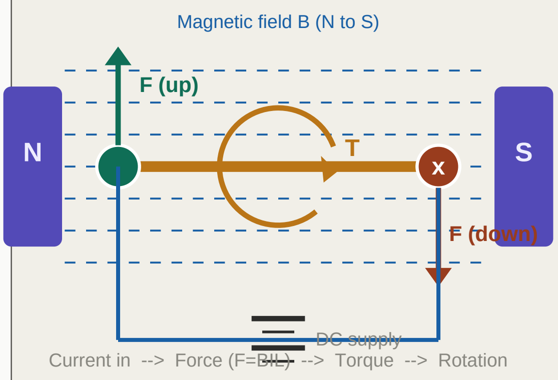

Fig 3: DC Motor — force on current-carrying conductors

What to write:

- Principle: F = BIL — conductor in B-field gets force
- Rule: Fleming's LEFT-Hand Rule gives direction of force
- Two conductors get OPPOSITE forces -> creates torque
- Commutator reverses current each half turn -> cont. torque
- Back EMF (Eb) = induced EMF opposing supply voltage
- Back EMF formula: Eb = V - Ia x Ra
- Self-regulation: load up -> speed down -> Eb down -> Ia up -> T up
- Formula: T = (P x phi x Z x Ia) / (2 x pi x A)

T proportional to phi x Ia | Eb = V - Ia x Ra

3 Lines That Always Get Marks — Write These First!

Generator: Mechanical -> Electrical | Right-Hand Rule | E = PphiNZ/60A | Commutator = AC to DC

Motor: Electrical -> Mechanical | Left-Hand Rule | T proportional to phi.Ia | Back EMF = V - IaRa

Commutator: Generator = converts AC to DC at output | Motor = maintains continuous unidirectional torque

Exam Tip: Always draw the diagram first, write principle, state the rule, give the formula. Done!